

One place for all your product content.

PIM + Social Campaign Module
ASP.NET Core · React · Python AI

Today, a single product has 6+ document types stored in 6+ different places:

📄 PI Sheet

📐 Line Drawing

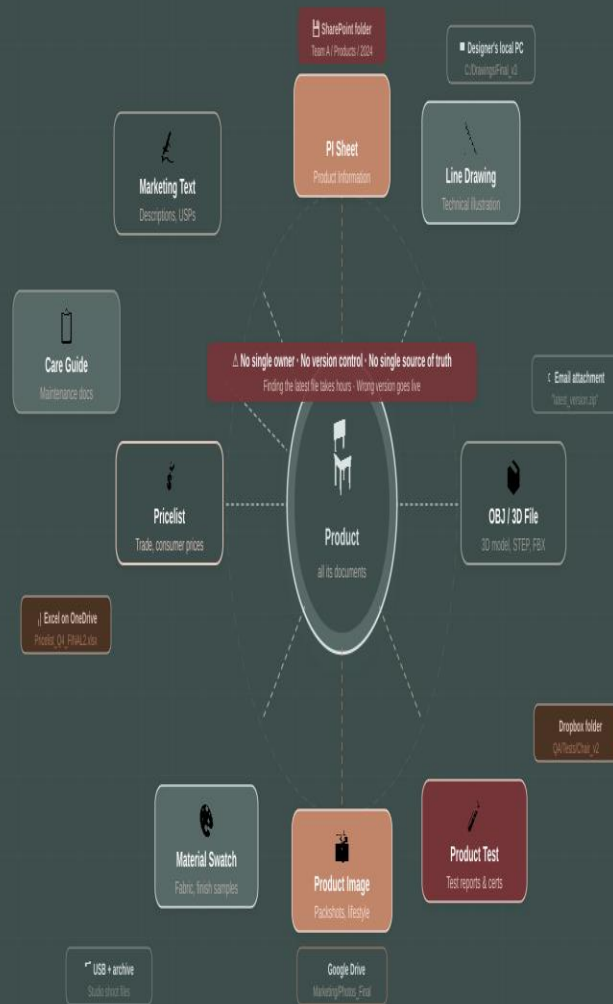
📦 OBJ / 3D File

🧪 Product Test

🖼️ Product Image

🎨 Swatch / Material

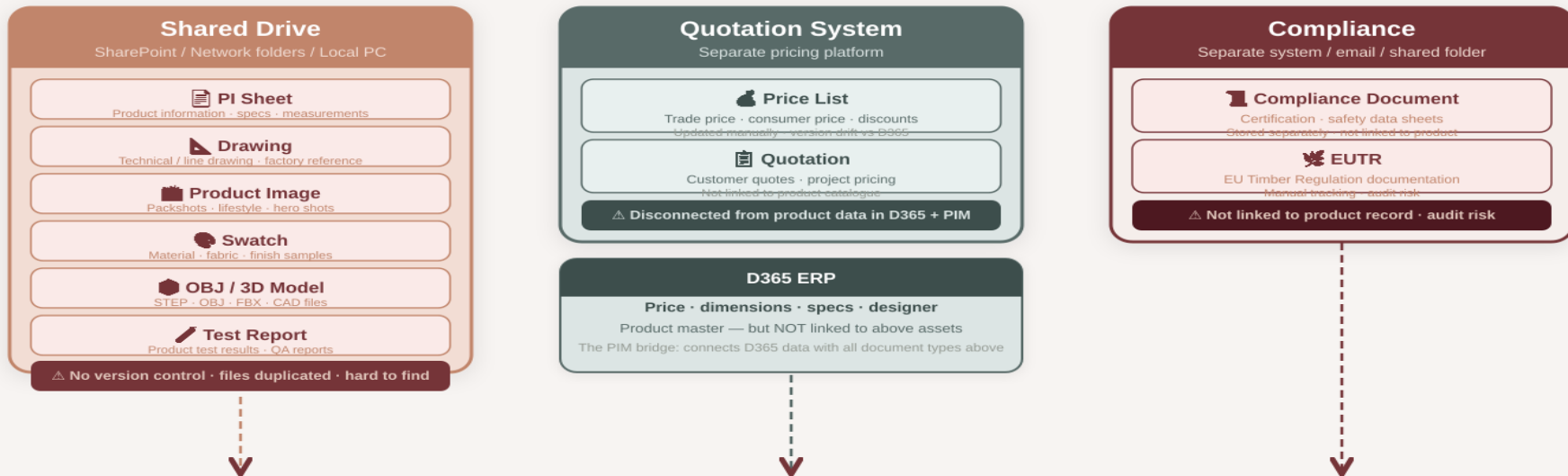
→ No version control · No single owner · Hours wasted searching



The Problem

Documents scattered across 3+ systems — no link to product in D365

WHERE DOES YOUR PRODUCT CONTENT LIVE TODAY?



RESULT: Documents scattered across 3+ systems — no single source of truth

Wrong PI Sheet version sent to customer · Missing compliance doc at audit · Price mismatch between quote and catalogue
Team wastes hours searching for the latest version of every document

3+

separate systems
for one product's docs

8+

hrs/week searching
for the right file version

0

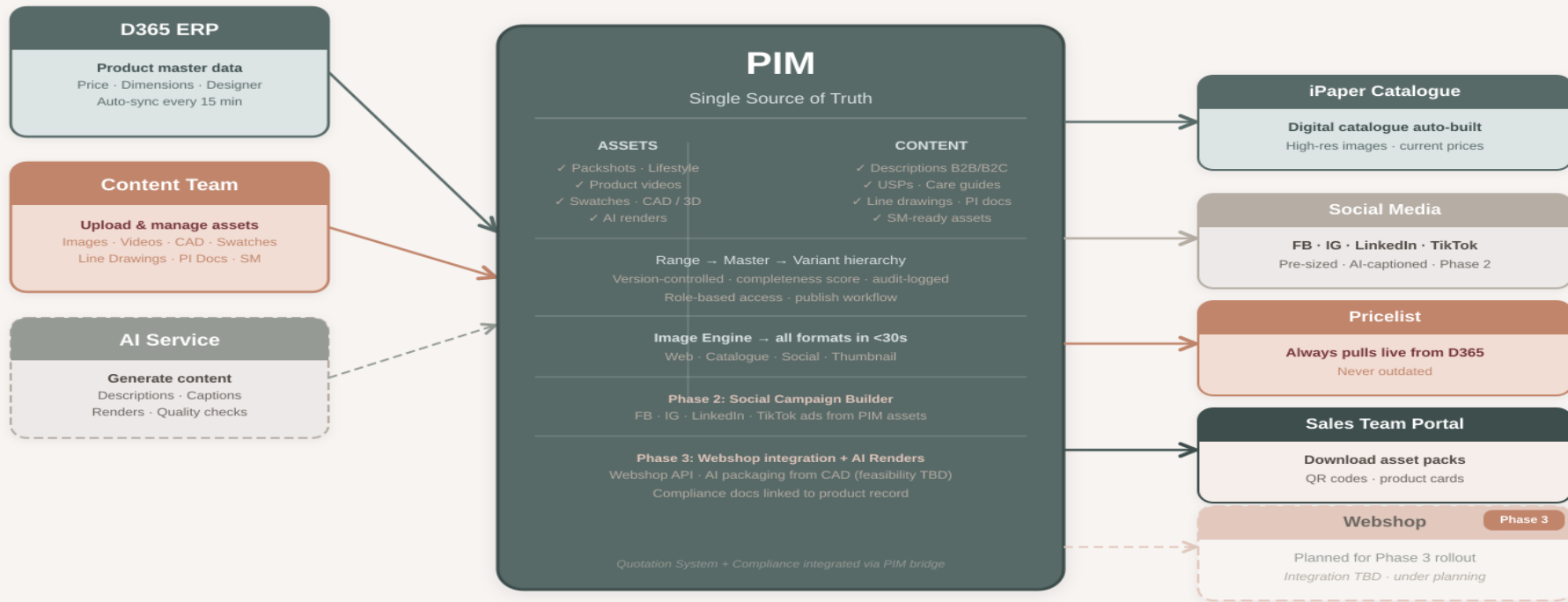
documents linked to product
Range / Master / Variant in D365

PIM solves this →

All documents linked to
Range / Master / Variant
One place - one version - always current

The Solution

PIM = one place for everything · push to every channel automatically



What content lives in PIM?

Every asset linked to Range / Master Number / Variant — structured like D365



Packshots

Hero · pack · lifestyle
all angles



Videos

Product · lifestyle
Reels · tutorials



Swatches

Material · color
fabric samples



CAD / 3D Files

STEP · OBJ · FBX
source for AI render



Line Drawings

Technical illustrations
for catalogue



PI Documents

Product info sheets
spec sheets · PDFs



AI Renders

Auto-generated from
CAD
all material variants



SM Assets

Social-ready formats
FB · IG · LI · TT



Text Content

Descriptions · USPs
Care guides (AI-
drafted)



Price Data

From D365 · auto-sync
always current



Lifestyle Scenes

Room shots · mood
boards
lifestyle contexts

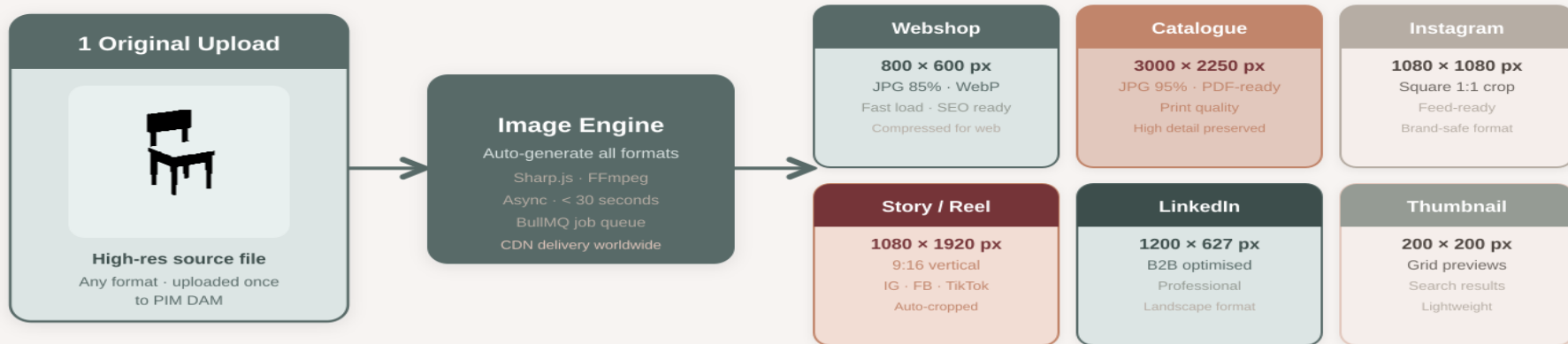


360° Spin Sets

Interactive views
for webshop (Phase 3)

Image Engine — Upload once, get every size

No more manual resizing · Auto-generate all formats for web, print and social in < 30 seconds



BEFORE PIM

Content team manually resizes every image for every channel

2–3 hours per product · error-prone · inconsistent results
Different person does it each time → different crops, quality

AFTER PIM — Image Engine

Upload once → all 6 formats generated automatically

In under 30 seconds · stored in CDN · pushed to all channels
Always consistent · always the latest version

6

formats generated
from 1 upload

<30s

processing time
async · non-blocking

CDN

global delivery
cache + invalidate on update

0

manual resizing
team time saved

Upload original → Image Engine processes → 6 formats in CDN → All channels updated automatically

Webshop · iPaper catalogue · Instagram feed · Story/Reel · LinkedIn · Thumbnail — all from one source file

Sharp.js (images) · FFmpeg (video) · BullMQ async job queue · Azure Blob Storage · CDN cache invalidation

How it works — 7 steps

From D365 to published — automated, traceable, fast

1

D365 Syncs

Product data every 15 min

2

Upload Assets

Images, video, CAD to DAM

3

Image Engine

Auto-resize for all channels

4

AI Writes Text

Descriptions, USPs, care

5

Review & Approve

Human always reviews first

6

Completeness

Score check before publish

7

Publish

iPaper · Social · Pricelist

Social Campaign Builder

Pick PIM assets · AI writes captions · Schedule and publish to all platforms in one flow

PIM — Campaign Builder

[+ New Campaign](#)**1**

Basics

2

Select Assets

3

AI Copy

4

Schedule

PIM Asset Library

IG / FB ✓



Chair_01.jpg

TT / Reel ✓



Sofa_life.mp4

LI / FB ✓



Swatch_linen.png



Table_01.jpg



Lamp_hero.jpg



Bed_lifestyle.jpg

AI Render ✓



Linen_AI_render.png

Line Drawing ✓



Chair_techline.pdf

Drag assets to campaign canvas →

Campaign Canvas — "Summer 2025 Launch"

FB

IG

LI

TT

Facebook Ad Preview · 1200×628



Chair_01.jpg · auto-sized to 1200×628

Aria Collection

AI caption loading...

Summer Promo CTA

[Shop Now](#)

Instagram Preview · 1080×1080



Auto-cropped to 1:1 square

AI writing caption... #furniture #design

AI is generating 3 caption variants for each platform...

B2B vs B2C tone · Platform-appropriate length · Hashtag suggestions · CTA recommendation

Variant A: Elegant & professional

Variant B: Playful & aspirational

Variant C: Direct + discount CTA

Asset Link Map — Know exactly where every asset is

See which campaigns, platforms, and channels use each asset — with live performance data

Media Hub — Asset Link Map

Where is each asset being used right now?

342

Total media assets in PIM

128

Actively linked to campaigns

89

Unused — ready to activate

14

Active campaigns running

Asset	Product	Linked to (Platform / Campaign)	Performance	Status
 aria_chair_lifestyle_v2.mp4 MP4 · 1080×1920 · 18 seconds · 24 MB Uploaded 3 weeks ago · quality score 94/100 Aria Collection	FB Summer 2025 Reel IG Q3 Story Campaign	✓ ✓	12.4K views CTR 3.2% · Eng. rate 6.8% Spend: \$248 this month	Active
 como_table_packshot_01.jpg JPG · 1200×628 · 2.1 MB · quality 91/100 Como Range	LI B2B Trade Post — October	Scheduled	— Not yet live	Sched.
 material_swatch_linen_cream.png PNG · 1080×1080 · 0.8 MB · quality 97/100 All Ranges	FB Material Spotlight Campaign IG Material Spotlight Campaign TT Trend Collab — Aug	✓ ✓ ✓	98K reach Eng. rate 8.1% Best-performing swatch asset	Active
 nova_sofa_lifestyle_sun.jpg JPG · 1200×800 · 3.4 MB — uploaded 45 days ago Nova Series	⚠ Not linked to any campaign or channel		Use in campaign →	Unused

Every asset shows exactly where it is used · Live performance data pulled every 6 hours from FB / IG / LI / TT

Unused assets flagged automatically · AI recommends top-performing assets for the next campaign

Top performer this month:

Swatch video — 2.4× avg CTR across IG Reels

89 assets never used in campaigns

AI can suggest best match for next campaign →

Analytics cron: every 6 hours

FB · IG Insights · LinkedIn · TikTok APIs

AI does the heavy lifting

Content team focuses on review + approval — not typing from scratch

AI Features — Across Every Phase

Python FastAPI · Claude API (Anthropic) · Always human-reviewed before publish

Phase 1 — Foundation

Product Descriptions

B2B + B2C from D365 metadata
Auto-draft on product creation
Editable in rich text editor

USPs & Care Guides

5 USP bullet points per segment
Material-based care instructions
Reusable per material type

Image Quality Check

Blur · resolution · background
Score shown before publish
Blocks low-quality from going live

Semantic Search Ready

Embeddings indexed silently
Powers Phase 2 smart search

Phase 2 — Growth

Social Caption Generation

3 variants per platform (FB/IG/LI/TT)
Hashtags + CTA · B2B or B2C tone
Editable before publish

Auto Image Tagging

Color · material · style · room type
Enables semantic search
Claude Vision API on upload

Asset Recommendation

Ranks best performers from history
"Swatch video 2x CTR on IG Reel"
Shown in Campaign Builder

Translation (5 locales)

EN / DE / FR / DK — brand glossary
Human review always required

Phase 3 — Vision

3D → Packshot (Research)

Feasibility study required first
Blender API · GPU pipeline
Not confirmed for delivery

Lifestyle Scene Gen.

If AI rendering confirmed
Stable Diffusion · ControlNet
Pending feasibility study

CTR Prediction

Predict ad performance before spend
ML model on 6-month history
Confidence interval shown

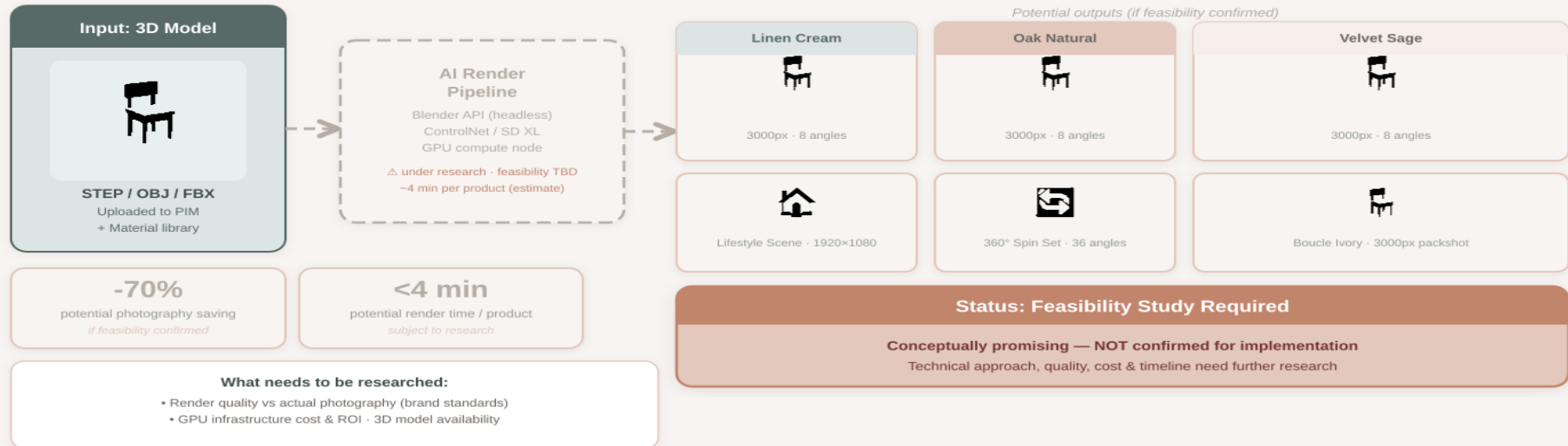
Optimal Posting Time AI

Best hour to post per platform
Based on 6-month audience data

AI Rendering — Future Vision Under Research

Conceptually promising · feasibility study required before committing to implementation

▲ FUTURE VISION — REQUIRES FEASIBILITY STUDY



Roadmap Position

Phase 1 & 2 → Confirmed

PIM Core + Social Campaign
Full delivery planned
ASP.NET Core · React · Python AI
Webshop integration in Phase 3

Phase 3 → Under Evaluation

AI Rendering feasibility study
Webshop API integration
Decision after Phase 2 is live
Not committed — research first

If Approved → Phase 3+

Pilot with 1 product range first
Side-by-side quality validation
Photography workflow in parallel
No studio booking replaced until proven

Tech Stack — built to scale

Three independent services · each deploys and scales on its own

ASP.NET Core 8

Backend API

- › REST endpoints + Hangfire jobs
- › D365 sync · Platform publish
- › Publish to iPaper / Pricelist
- › PostgreSQL + Redis + Typesense

React 18 + TypeScript

Frontend

- › Admin UI · Media Hub
- › Campaign Builder (drag-drop)
- › Asset Link Map dashboard
- › Analytics with Recharts

Python FastAPI

AI Service

- › Claude API (Anthropic)
- › Text · Caption · Quality check
- › Image tagging · Embeddings
- › ML + Render pipeline (Phase 3)

Shared: Azure Blob + CDN · RabbitMQ · TimescaleDB · Redis · Docker / Kubernetes · GitHub Actions CI/CD

3-Phase Roadmap

Start small, deliver value fast, grow over time

1

Phase 1 — Foundation

3–4 months · 3–4 developers

Deliverables:

- ✓ D365 product sync (every 15 min)
- ✓ DAM — upload & manage all assets
- ✓ Image Engine — auto-resize all formats
- ✓ AI text drafts (descriptions, USPs)
- ✓ Publish → Webshop + iPaper
- ✓ Admin UI + role-based access
- ✓ Completeness score per channel
- ✓ D365 price/spec change notifications
- ✓ Audit trail + version history

Business Impact

Zero shared drive uploads
Save 8+ hours/week per content manager
Correct content on webshop, always

Tech Stack

ASP.NET Core 8 · React 18 + TypeScript
Python FastAPI (AI Service)
PostgreSQL · Redis · Typesense · RabbitMQ
Azure Blob + CDN · Docker / Kubernetes

2

Phase 2 — Growth

4–5 months · 4–5 developers

Deliverables:

- ✓ Social Media Hub + platform tagging
- ✓ Campaign Builder (drag-drop canvas)
- ✓ AI caption + hashtag generation
- ✓ Publish → FB / IG / LinkedIn / TikTok
- ✓ Asset Link Map dashboard
- ✓ Analytics dashboard (6h pull)
- ✓ Semantic search + auto image tagging
- ✓ AI asset recommendation engine
- ✓ Multi-locale translation (AI-assisted)

Business Impact

All social campaigns built from PIM
100% visibility — where every asset lives
Zero manual caption writing for social ads

New tech added

Meta Marketing API v19 · LinkedIn API v2
TikTok Marketing API v2
TimescaleDB · FullCalendar · Recharts
FFmpeg video transcode · Sharp social formats

3

Phase 3 — Vision

6+ months · 5–6 developers

Deliverables:

- ✈ AI rendering from 3D CAD
- ✈ Material & color variant auto-renders
- ✈ Lifestyle scene generation (AI inpaint)
- ✈ 360° spin set for interactive webshop
- ✈ CTR prediction before ad spend
- ✈ Optimal posting time AI
- ✈ Full localization (5 languages)
- ✈ Headless CMS / GraphQL API
- ✈ Multi-brand / multi-tenant support

Business Impact

-70% photography cost per variant
CTR prediction accuracy 65%+
All 5 locales covered, AI-quality text

New tech added

Blender API (headless) · ControlNet
Stable Diffusion XL · ComfyUI pipeline
MLflow model registry · GPU node pool
Hot Chocolate GraphQL · PostgreSQL RLS

Why build this now?



Time wasted

Content team spends 8+ hours/week searching for the right file version. Every hour saved = direct cost reduction.



Wrong content live

Outdated PI sheet, wrong drawing version, incorrect image go live. PIM enforces completeness score before publish.



AI is ready now

Claude API generates production-quality product descriptions and captions today. Starting early = competitive advantage.



Social = growth

100% of campaign assets from PIM. No local copies. Asset Link Map shows what's working in real-time.

Next Steps

1

Stakeholder sign-off

Review this deck · Confirm scope · Approve Phase 1 budget

2

Technical spikes (Week 1–2)

Validate D365 API access · Validate Meta API sandbox with test ad account

3

Environment setup (Week 2)

Docker Compose: ASP.NET Core + React + Python + PostgreSQL + Redis + RabbitMQ

4

Sprint 1 kick-off (Week 3)

Repo · CI/CD pipeline · DB schema migrations · API scaffold

5

Phase 1 launch (Month 3–4)

Content team live · Zero shared drives · iPaper pulls from PIM

Questions? Let's discuss → Product Manager / Business Analyst / Technical Manager